

CSC-335

HCI & Computer Graphics

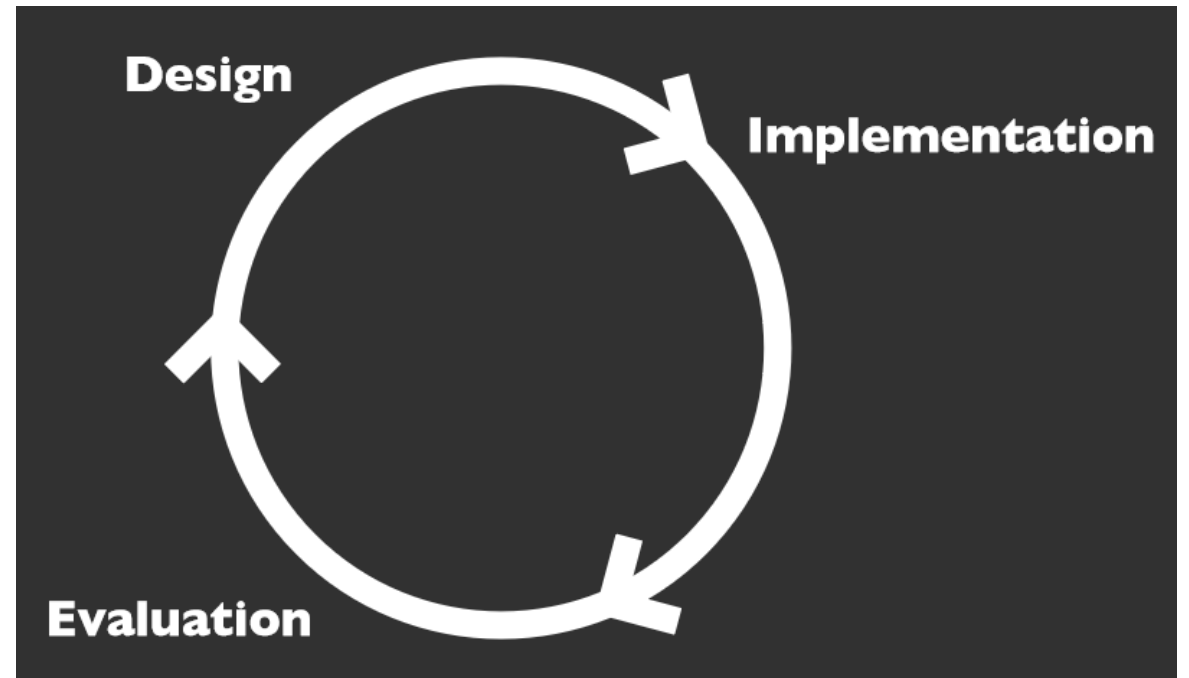
Lec 1: Introduction

Human Computer Interaction

- What is HCI?

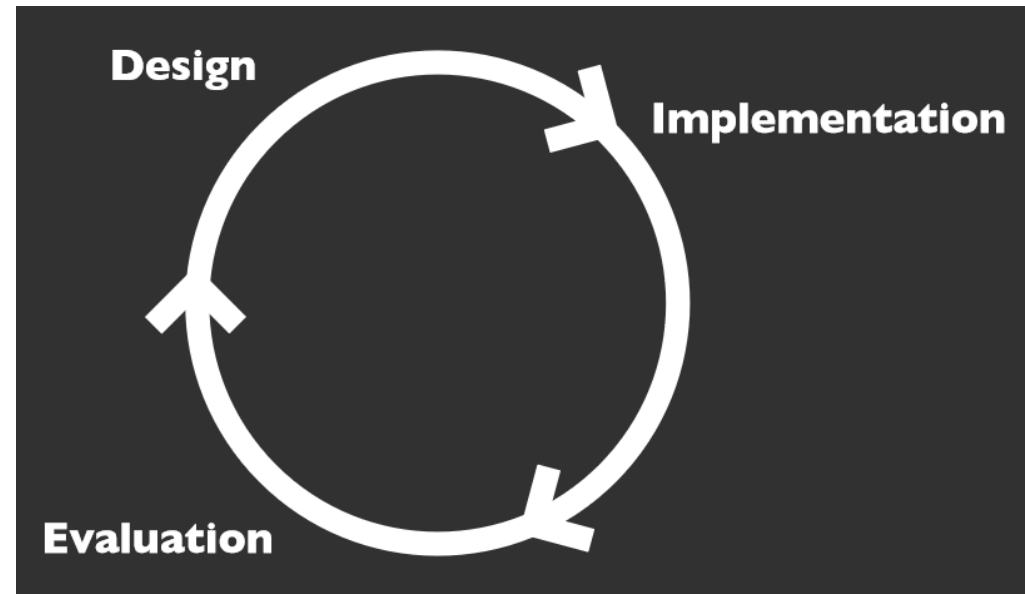
Human Computer Interaction

- What is HCI?
- Working in quick iterations so we can figure out problems sooner rather than later



Human Computer Interaction

- What is HCI?
- What makes a design a good design?
- What makes it a bad one?
- What is one factor which is central to all interaction design?

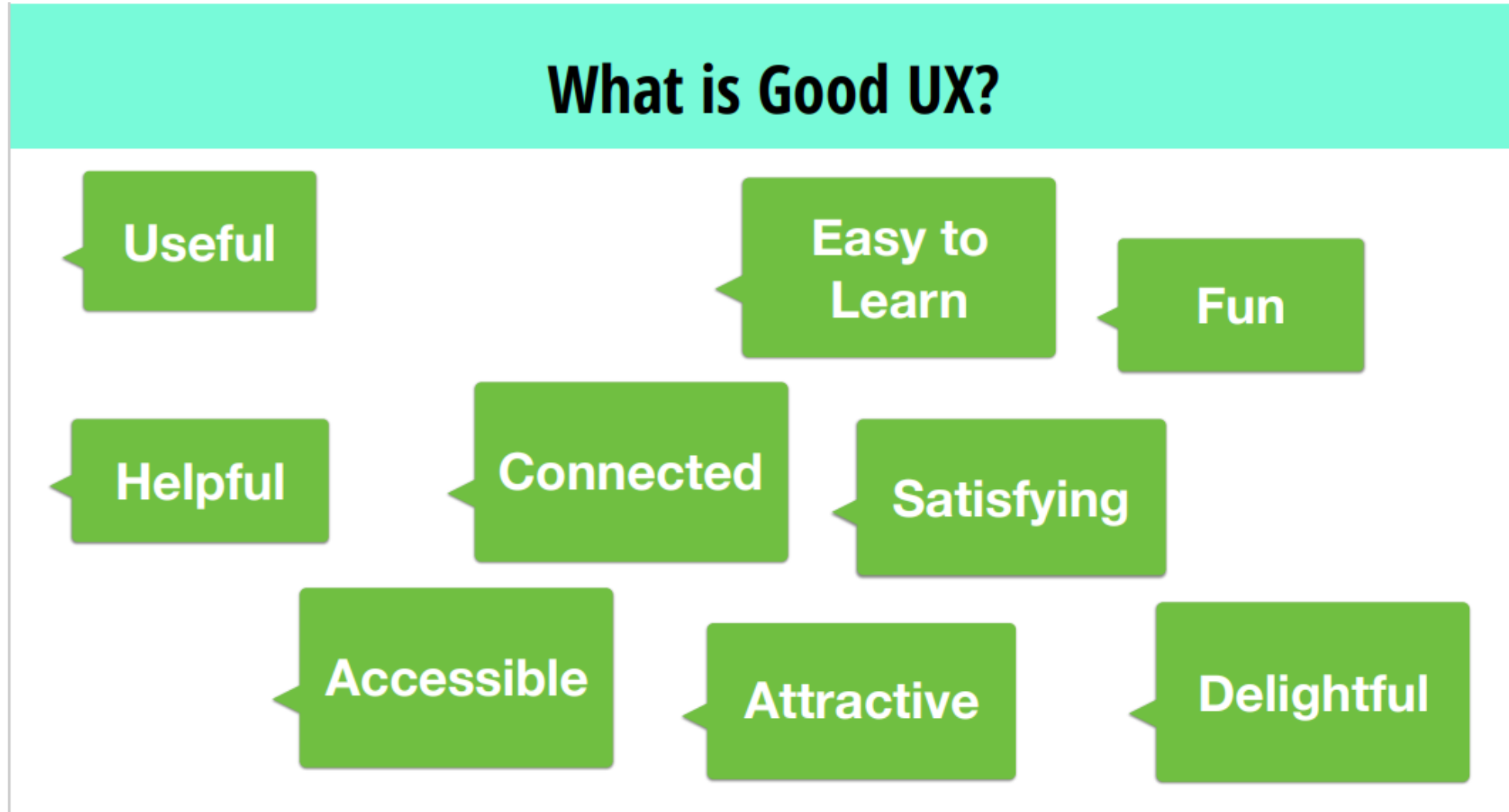


User Centered Design

- Design with people in mind; people's tasks, goals, and values should drive your development
- Work with users throughout the process
- Assess decisions from the users' point of view, their work and their environment
- Pay attention to people's abilities and situations

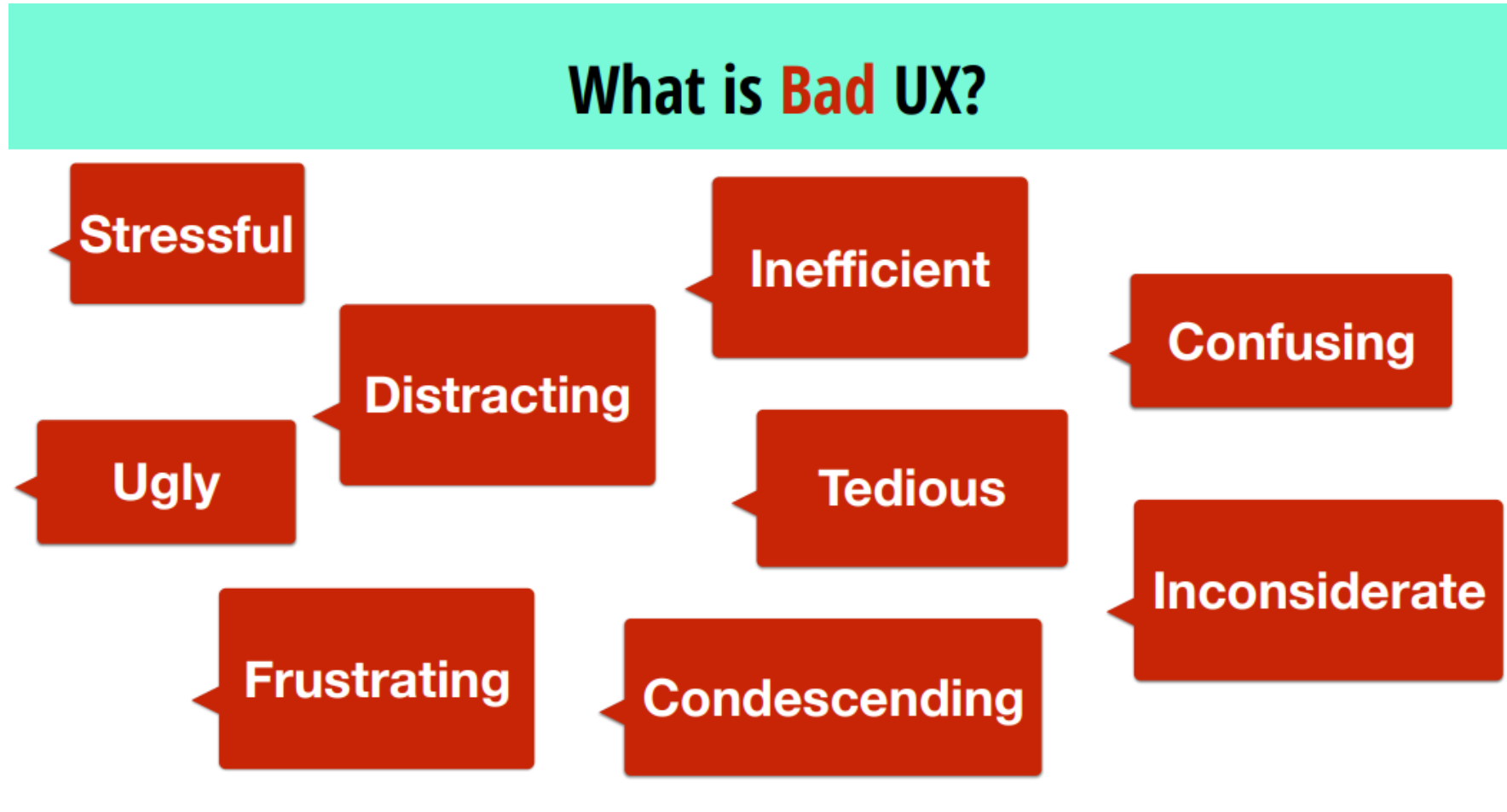
What is a good User Experience?

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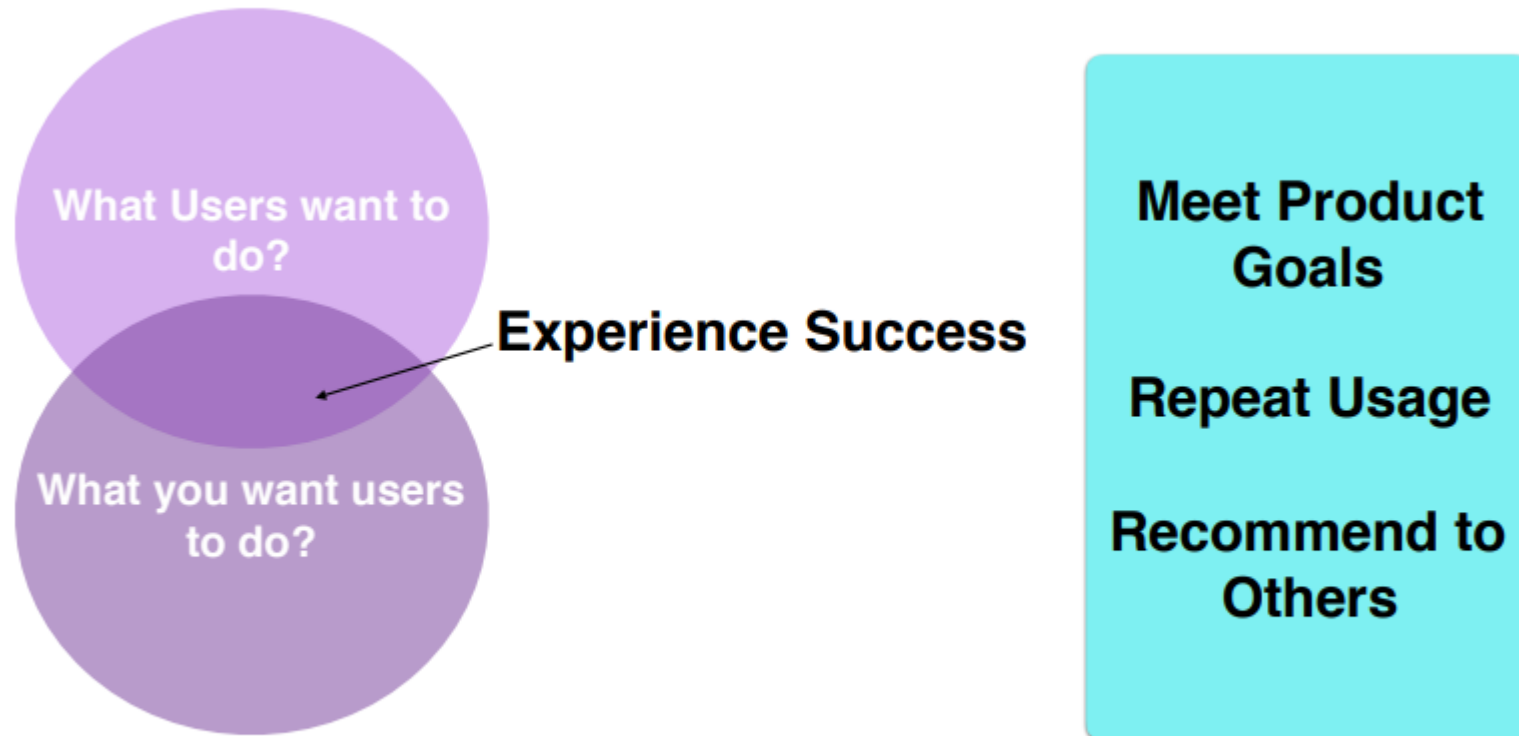
What is a bad User Experience?

What is a bad User Experience?



Why does it matter?

Why does it matter?

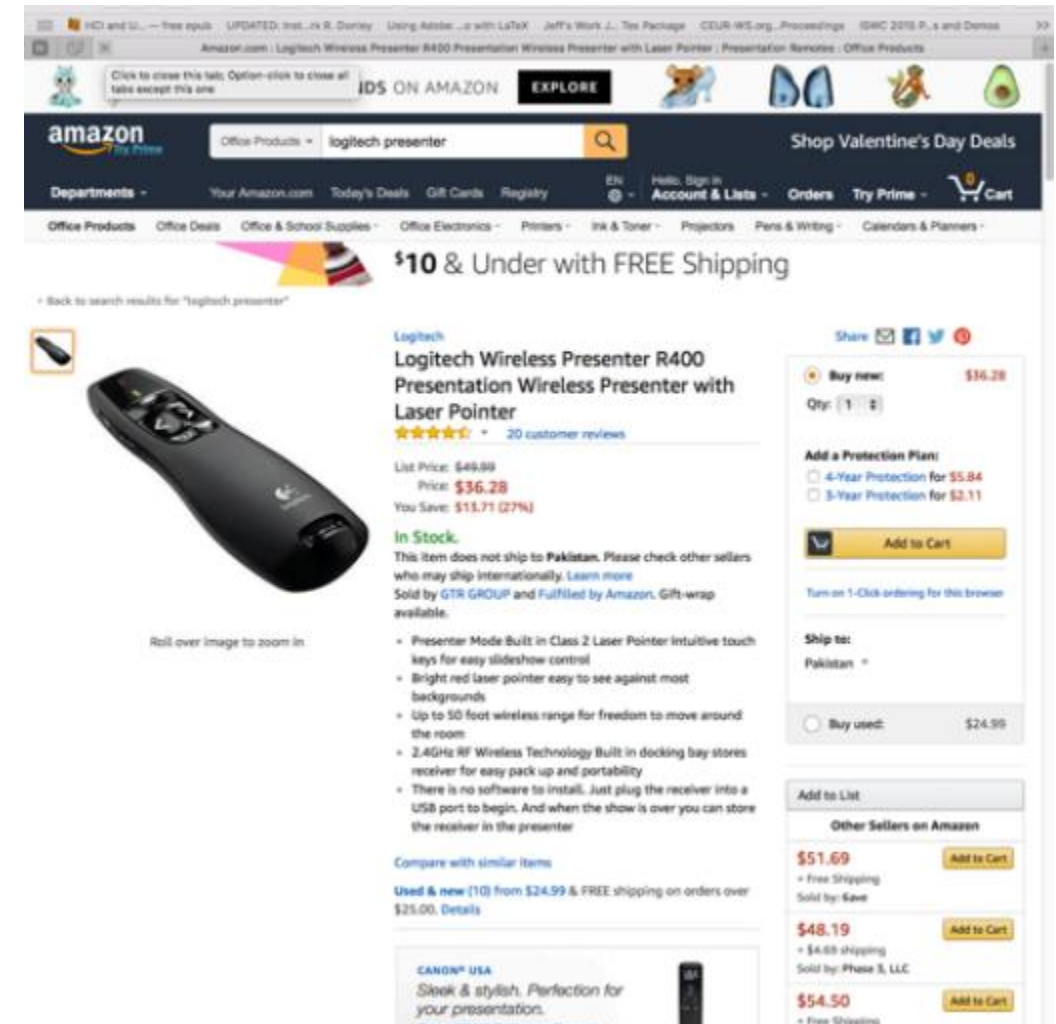


The UX as a whole?

- UX is the experience people have when they interact with your product
- Using the product
- Choosing it
- Acquiring
- Learning to use it
- Fixing the product
- Upgrading the product

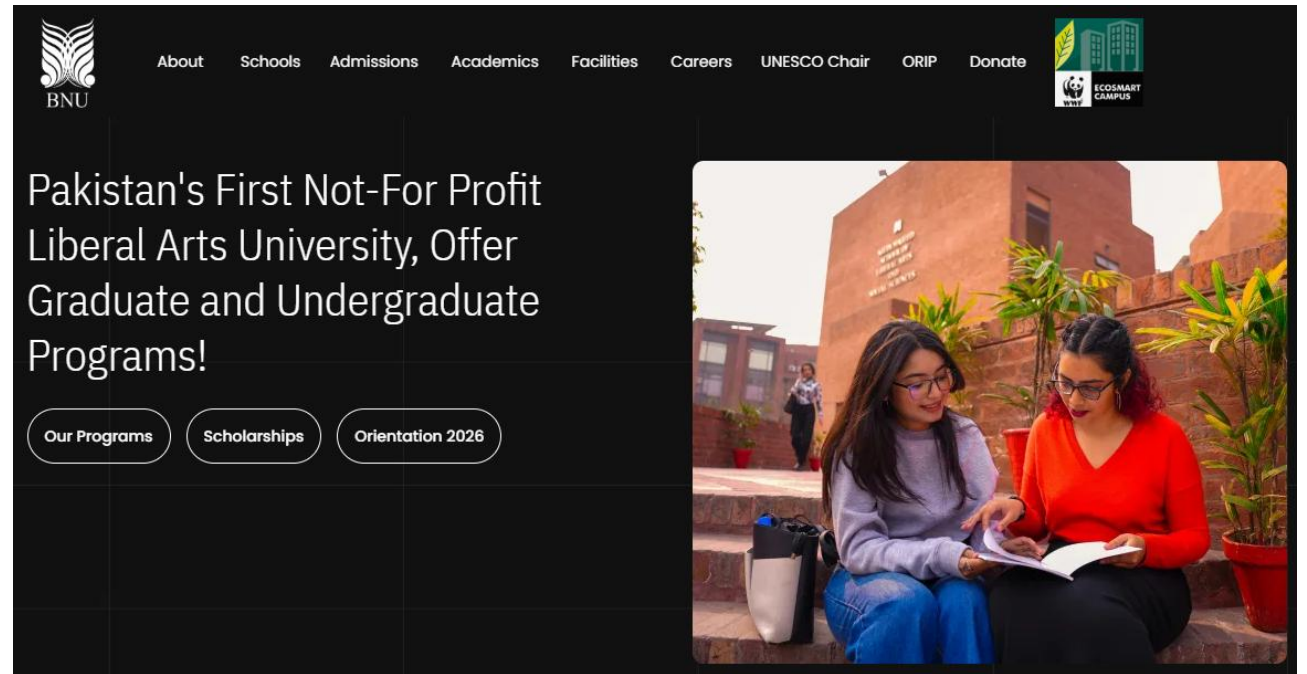
Example - 1

- Amazon.com
- Buy Stuff
- Enjoy Stuff
- Come back again
- Recommend to others



Example - 2

- Bnu.edu.pk
- Find information
- Understand Programs
- Streamline admission services
- Recommend to Others



Example - 3

- Glucometer
- Improve health
- Do no harm
- Convince providers
- Recommend to others



How UX Relates to Interaction Design and Other Fields



Interaction Design

- What is Interaction Design?

- Develop interactive products that are usable
- Easy to learn
- Effective to use
- Enjoyable user experience

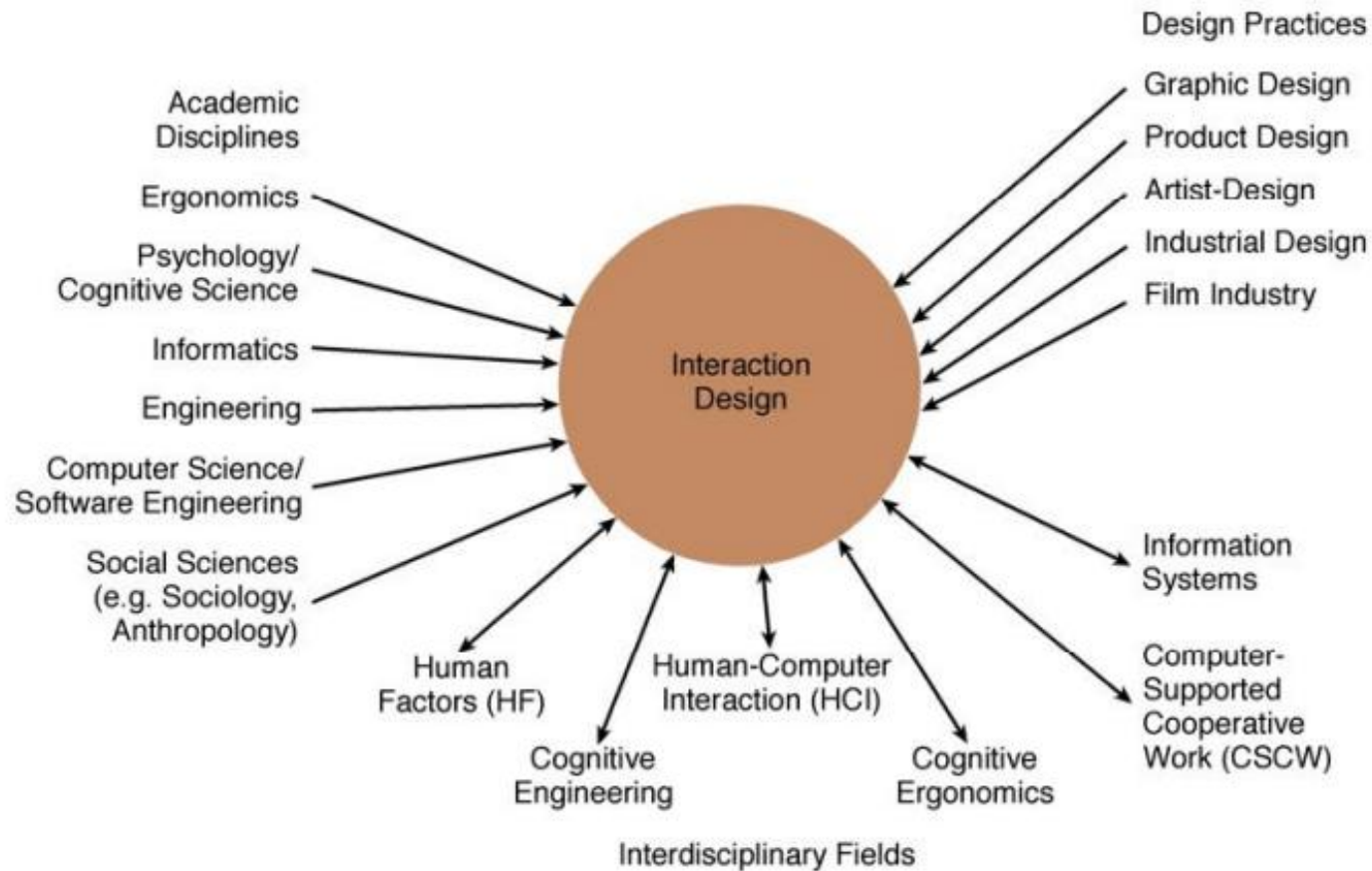
- Goal of Interaction Design

- Develop usable products
 - Usability means easy to learn, effective to use and provide an enjoyable experience
- Involve users in the design process

Which Kind of Design?

- Number of other terms used emphasizing what is being designed, e.g.
 - user interface design, software design, user-centered design, product design, web design, experience design (UX)
- Interaction design is the umbrella term covering all of these aspects
 - fundamental to all disciplines, fields, and approaches concerned with researching and designing computer-based systems for people

HCI & Interaction Design



What do professionals do in the ID business?

- interaction designers - people involved in the design of all the interactive aspects of a product
- usability engineers - people who focus on evaluating products, using usability methods and principles
- web designers - people who develop and create the visual design of websites, such as layouts
- information architects - people who come up with ideas of how to plan and structure interactive products
- user experience designers (UX) - people who do all the above but who may also carry out field studies to inform the design of products



UX in the Light of ID

User Experience

- How a product behaves and is used by people in the real world
- The way people feel about it and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it



“every product that is used by someone has a user experience: newspapers, ketchup bottles, reclining armchairs, cardigan sweaters.” (Garrett, 2010)

*“Great design is the
iteration of good
design”*



Cannot design a user
experience, only
design for a user
experience





Usability vs. User Experience

Usability vs. User Experience

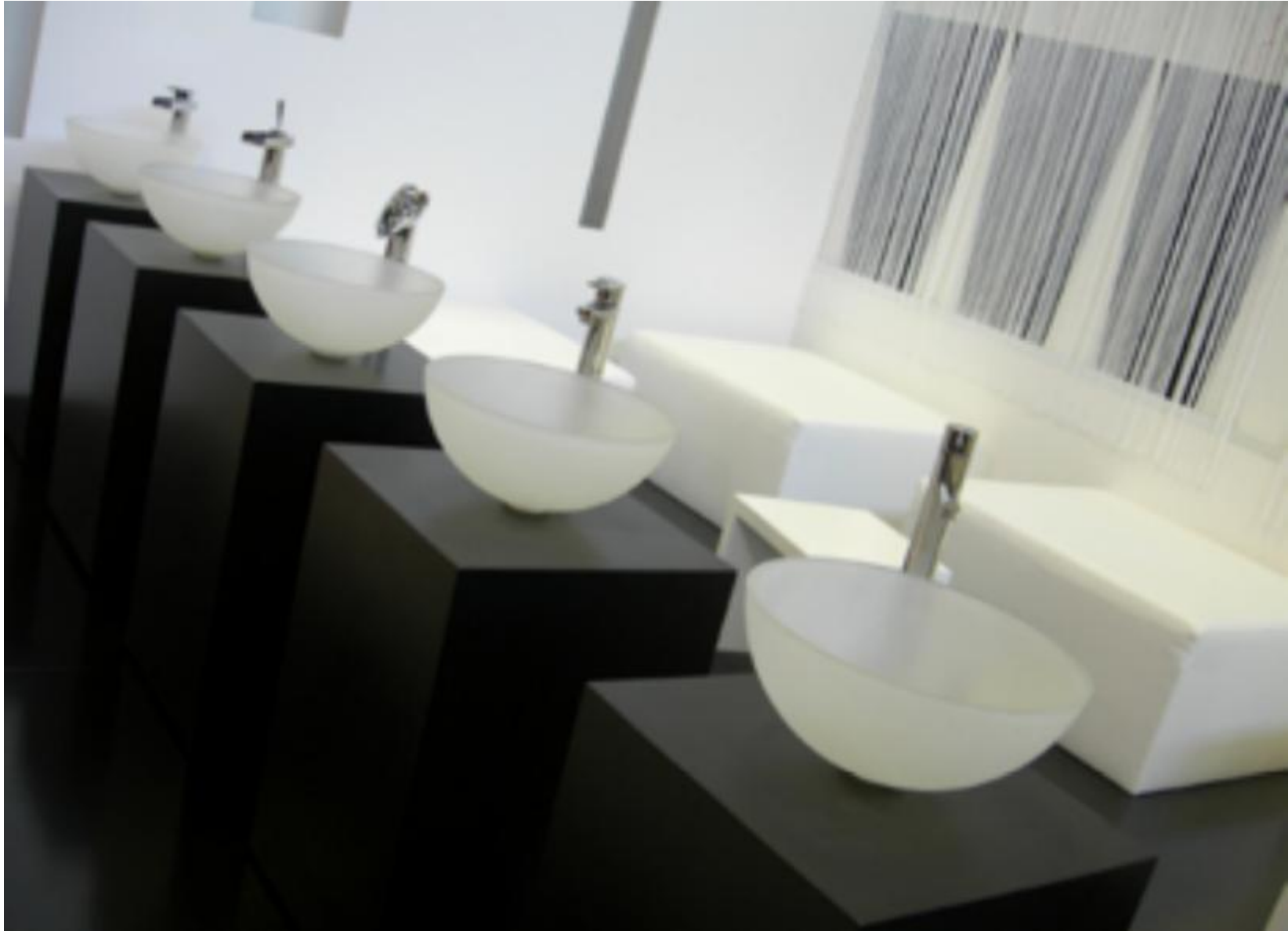
- “Usability is the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.” - Definition of Usability DIN EN ISO 9241-11

Is Usability enough?

- Fulfilling this definition leads to:
 - Products that do not prevent users from performing their tasks
 - Products that can be used effectively and efficiently
 - Products that do not annoy users
 - Is this really sufficient to build great products that users enjoy to use?









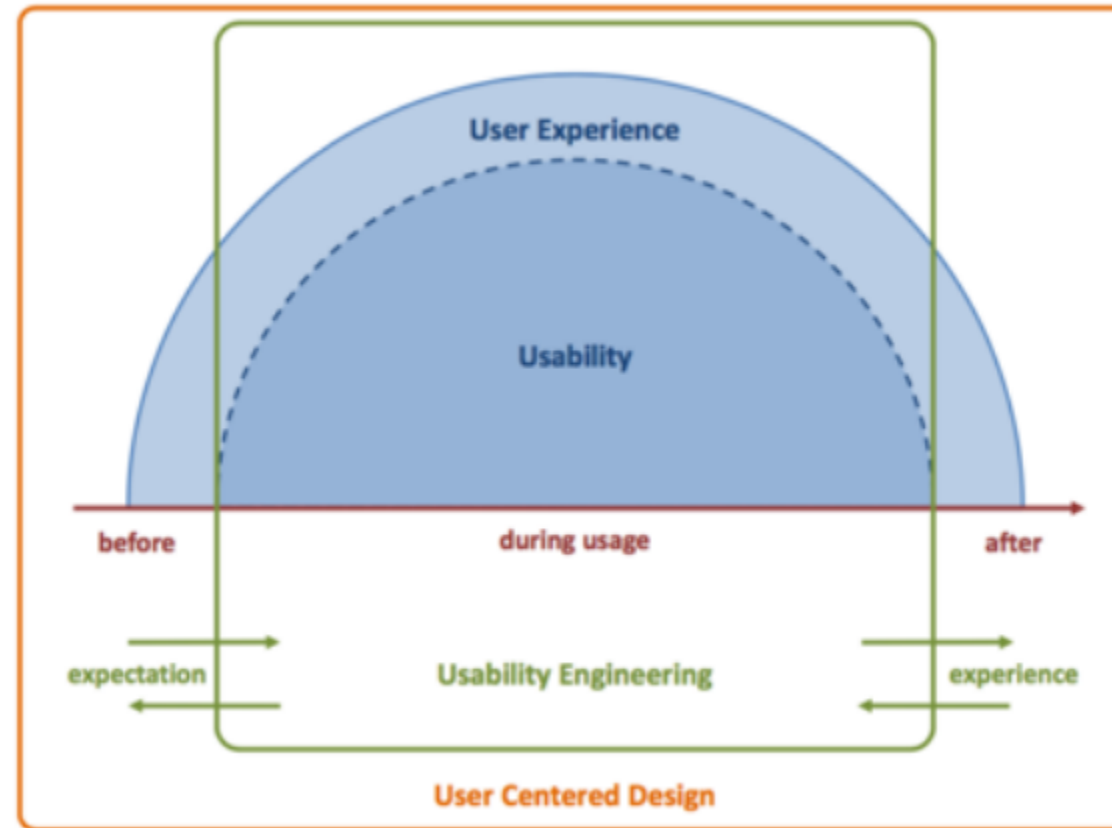
User Experience is more than just Usability!

- Product related User Experience Factors
- User Experience is influenced by a lot of different factors
- Some of them are actual product related factors like
 - features or quality attributes
 - Other factors concern: the brand, the image, the packaging, marketing
 - Can you think of any other factors?

User Experience - Formally Defined!

- “User experience includes all aspects of the user’s experience when interacting with the product, service, environment or facility.
- It is a consequence of the presentation, functionality, system performance, interactive behavior, and assistive capabilities of the interactive system.
- It includes all aspects of usability and desirability of a product, system or service from the user’s perspective.”

User Centered Design - Relationships



Usability & User Experience Goals



Usability Goals

- Effective to use
- Efficient to use
- Safe to use
- Have good utility
- Easy to learn
- Easy to remember how to use

User experience goals

- Desirable Aspects
- Undesirable aspects

User experience goals

- Desirable Aspects
 - satisfying helpful fun
 - enjoyable motivating provocative
 - engaging challenging surprising
 - pleasurable enhancing sociability rewarding
 - exciting supporting creativity emotionally fulfilling
 - entertaining cognitively stimulating
- Undesirable aspects
 - boring unpleasant
 - frustrating patronizing
 - making one feel guilty making one feel stupid
 - annoying cutesy
 - childish gimmicky

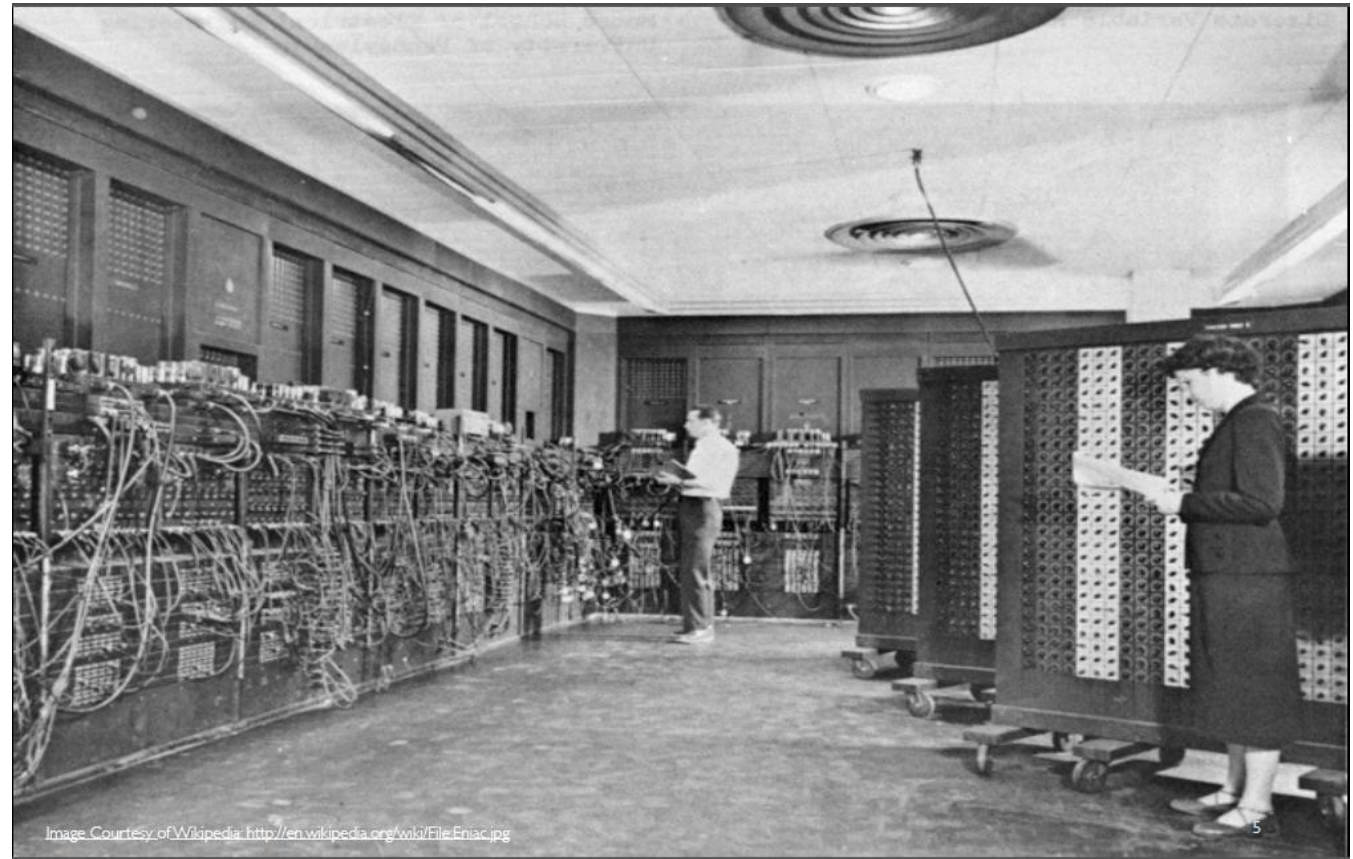
Balancing Usability and user experience goals

- Selecting terms to convey a person's feelings, emotions, etc., can help designers understand the multifaceted nature of the user experience
- How do usability goals differ from user experience goals?
- Are there trade-offs between the two kinds of goals?
 - e.g. can a product be both fun and safe?
- How easy is it to measure usability versus user experience goals?

How has the way humans interact
with computers changed over
time?
&
why does that matter for UX/UXE
today?

Timeline - 1

- Early Computing aka Batch Processing
 - No Interaction
- Model: Submit -> wait -> get output
 - eg: Punch cards & Mainframes
- UX Issue:
 - No real-time feedback
 - No usability concept yet



Timeline - 2

- Command Line Interaction (CLI)
- Model: Text commands
 - eg: DOS, Linux terminal
- UX issues:
 - Expert users only
 - Memorization required
 - Low discoverability
 - Cognitive load is high
 - Learnability is low

```
Current date is Tue 1-01-1980
Enter new date:
Current time is 7:48:27.13
Enter new time:

The IBM Personal Computer DOS
Version 1.10 (C)Copyright IBM Corp 1981, 1982

A>dir/w
COMMAND COM FORMAT COM CHKDSK COM SYS COM DISKCOPY COM
DISKCOMP COM COMP COM EXE2BIN EXE MODE COM EDLIN COM
DEBUG COM LINK EXE BASIC COM BASICA COM ART BAS
SAMPLES BAS MORTGAGE BAS COLORBAR BAS CALENDAR BAS MUSIC BAS
DONKEY BAS CIRCLE BAS PIECHART BAS SPACE BAS BALL BAS
COMM BAS
26 File(s)
A>dir command.com
COMMAND COM 4959 5-07-82 12:00p
1 File(s)
A>
```

Timeline - 3

- GUI/WIMP Paradigm
- Model: GUI or Windows, Icons, Menus, Pointer
 - eg: Windows, MacOS, Desktop apps
- UX issues:
 - Visual interaction replaces memorization
 - Discoverability increases
 - Affordances matter
 - Direct manipulation begins
- Heuristics & usability principles emerge

Timeline - 4

- Web & Hyperlink Paradigm
- Model: Navigation-based interaction
 - eg: websites & browsers
- UX impact:
 - Shift from applications to information spaces
 - Information architecture becomes critical
 - Navigation matters
 - Consistency becomes key

Timeline - 5

- Touch & Mobile Paradigm
- Model: Direct touch manipulation
 - eg: smartphones & tablets
- Shift:
- UX consequences:
 - Pointer to finger
 - Target size matters
 - Gesture discoverability issues eg: swipe left to archive in Gmail
 - Context-aware UX
- Context-aware:
 - Location: map reccs
 - Time: apps show breakfast items in the morning
 - Behavior: most recently used apps shown first
 - Orientation: portrait vs landscape

Timeline - 6

- Voice & Conversational Interfaces
- Model: Natural language interaction
 - eg: Alexa, Siri, Chatbots
- UX challenges:
 - Interface disappears
 - No visual affordance: don't know what actions are possible
 - Feedback is invisible: eg: Voice assistant pauses for 3 seconds.
 - Error recovery is harder, no way of going back etc

Timeline - 7

- What timeline are we entering now?

Timeline - 7

- AI & Adaptive Interfaces (Current Shift)
- Model: System adapts to user
- eg: ChatGPT, AI copilots, recommendation systems
- UX implications:
 - User no longer fully controls flow
 - Trust becomes key
 - Transparency matters
 - Explainability needed
 - Ethical UX required

Timeline Summary

- Each paradigm shift changed:
 - How users think
 - What designers must optimize
 - What engineers must implement
- **UX rules are not static — they evolve with interaction paradigms**

Summary

- <https://www.youtube.com/watch?v=Ovj4hFeko7c>



- Any Questions?